Third System
AI Presence Index

What AI Presence Does Not Predict

Amazon Best Sellers Rank as discriminant validity evidence for the AIAS construct

May 2026 · Pablo Ulpiano González Castro

AIAS™ Measurement Program · Discriminant Validity

AI Presence scores show zero correlation with Amazon Best Sellers Rank across three consumer-goods categories. Brands that language models recognize are not the same brands that dominate retail sales. This is not a failure of the measure — it is evidence that AI Availability captures something genuinely new.

rho ≈ 0.000 Pooled correlation · **0 / 3** Substrates with signal · **104 / 106** Amazon coverage · **3 orders** BSR range

This study tested whether brands that score high on AI Presence also sell well on Amazon. They do not. Across kitchen knives, audiophile headphones, and skincare, the correlation between AI Presence Index (C_P) and Amazon Best Sellers Rank was indistinguishable from zero — not merely non-significant, but effectively absent.

The null result is the finding. Combined with the strong positive correlation between C_P and Google Trends search interest found in v0.25 (rho = 0.74), this study completes a convergent–discriminant validity pair: AI Presence tracks brand salience in AI knowledge systems (converges with search interest) but does not track commercial outcomes (diverges from retail sales rank).

For brand strategists, the implication is clear: AI Availability is not a proxy for market share. A brand can be universally recognized by language models and still rank poorly on Amazon, or dominate Amazon sales while being invisible to AI systems. These are independent dimensions of brand access — exactly what the Tri-System framework predicts.

What we measured

We correlated two signals across 106 brands in five consumer-goods categories.

AI Presence Index (C_P): How many of six major language models recognize each brand in its category (integer 0–6). Three substrates provided within-category variance: kitchen knives (26 brands, v0.16 retrofitted), audiophile headphones (16 brands, v0.19), and skincare (24 brands, v0.20). A fourth substrate — cosmetics (v0.21) — showed a ceiling effect where all 24 brands scored $C_P = 6$, providing a natural control. A fifth substrate (premium kitchenware, v0.17) was excluded because its Phase A data was incomplete.

Amazon Best Sellers Rank (BSR): The sales-velocity rank of each brand's best-selling product on Amazon, acquired via automated browser extraction in a single 48-hour window. Lower rank = more sales. Coverage: 104 of 106 brands listed (98.1%).

Test: Spearman rank correlation per substrate (does higher C_P predict lower/better BSR?), plus pooled cross-substrate correlation and a Cell A separation test.

FINDING 01

Zero correlation, uniformly

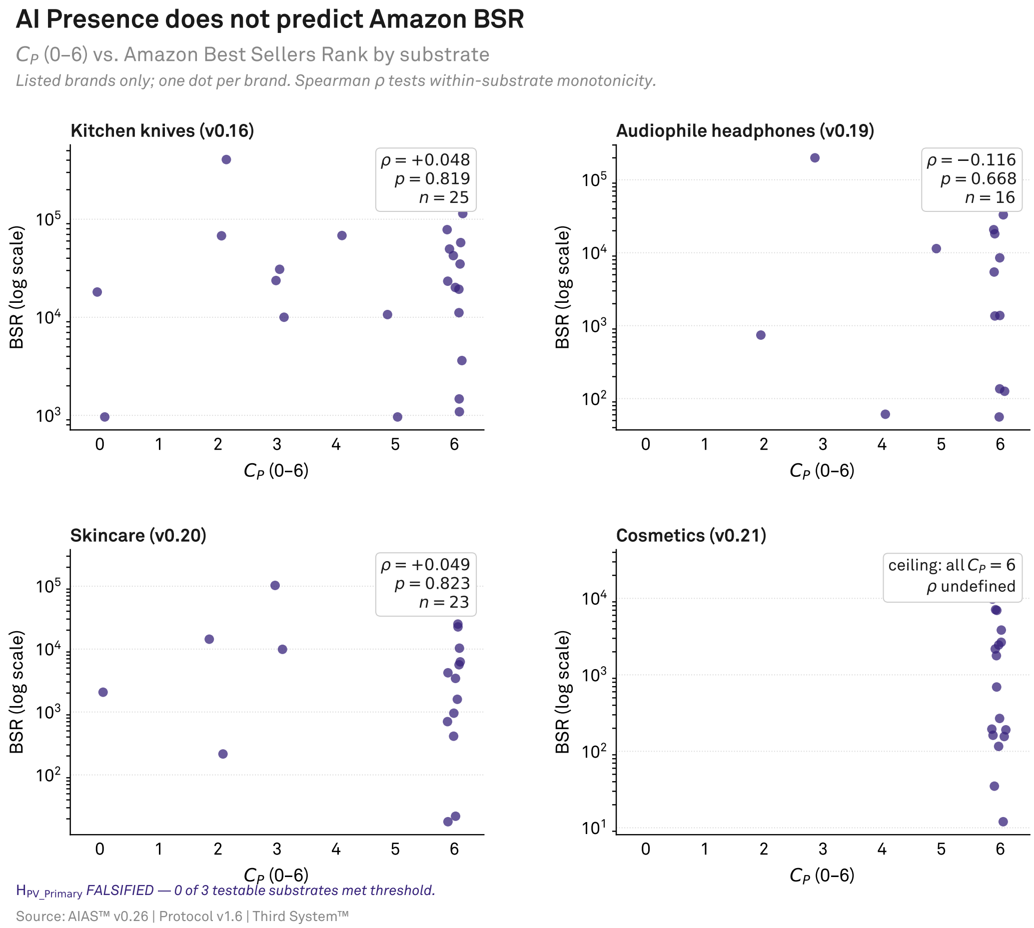


Figure 1 · AI Presence does not predict Amazon BSR. C_P (0–6) vs. Amazon Best Sellers Rank by substrate. Each panel shows listed brands with Spearman rho annotated. Kitchen knives, audiophile headphones, and skincare show indistinguishable-from-zero correlations; cosmetics shows the ceiling effect (all C_P = 6).

Within-substrate Spearman correlations: kitchen knives rho = +0.048 (p = 0.819, n = 25), audiophile headphones rho = −0.116 (p = 0.668, n = 16), skincare rho = +0.049 (p = 0.823, n = 23). Pooled across 88 brands: rho = −0.0002 (p = 0.998).

The correlations are not merely non-significant — they are effectively zero. Consider the kitchen knives substrate: Wüsthof (C_P = 6, BSR 11,138) and CCK Chan Chi Kee (C_P = 5, BSR 963) both score high on AI Presence, yet CCK outsells Wüsthof on Amazon by a factor of 11. Meanwhile Hengtai (C_P = 0, BSR 963) matches CCK's sales rank despite being invisible to every language model in the panel.

In audiophile headphones, Dan Clark Audio (C_P = 6, BSR 56) and Sony (C_P = 6, BSR 61) dominate Amazon sales, but so does Beyerdynamic (BSR 136) and Audio-Technica (BSR 126) — while Stax (C_P = 6, BSR 81,214) scores identically on AI Presence yet ranks 1,400 times lower on Amazon. AI Presence tells you nothing about where a brand sits in the retail sales hierarchy.

FINDING 02

The ceiling proves the point

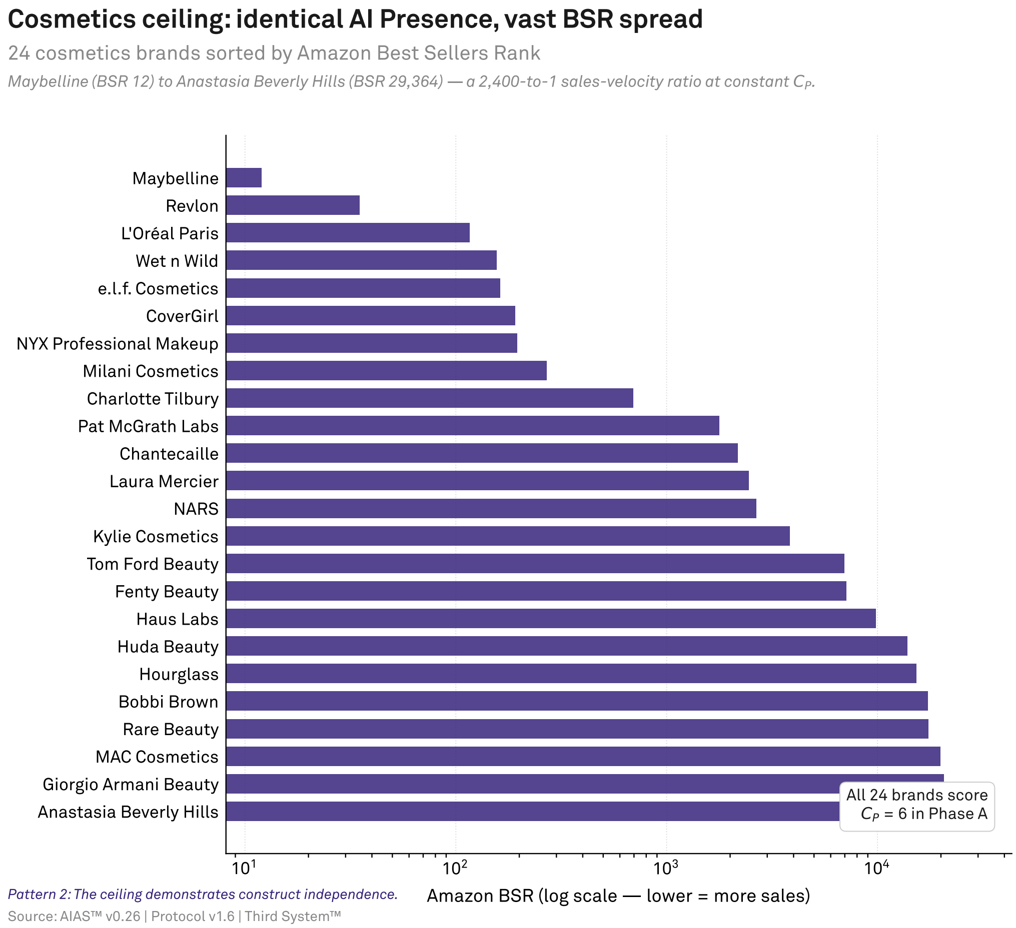


Figure 2 · The cosmetics ceiling. 24 cosmetics brands at C_P = 6 (full AI recognition), sorted by Amazon BSR. Mass-market brands (Maybelline BSR 12, Revlon 35, L'Oréal 116) dominate Amazon; prestige brands (Tom Ford 6,963, Anastasia Beverly Hills 29,364) rank orders of magnitude lower at identical AI Presence. Identical predictor, three-orders-of-magnitude outcome spread.

In cosmetics, all 24 brands achieved C_P = 6 (full AI recognition), yet BSR varied from 12 (Maybelline) to 29,364 (Anastasia Beverly Hills) — a 2,400-to-1 ratio in sales velocity with identical AI Presence scores.

The mass-market brands dominate Amazon: Maybelline (BSR 12), Revlon (35), L'Oréal Paris (116), Wet n Wild (156), e.l.f. Cosmetics (162), CoverGirl (191), NYX (195). Meanwhile, prestige brands that language models know equally well — Tom Ford Beauty (6,963), Giorgio Armani Beauty (20,698), Anastasia Beverly Hills (29,364) — rank orders of magnitude lower on Amazon.

This is a natural experiment: AI Presence is held constant at the maximum while retail performance varies by three orders of magnitude. The ceiling does not merely prevent within-substrate correlation. It demonstrates the independence of the two constructs. Full AI recognition is compatible with any level of retail performance.

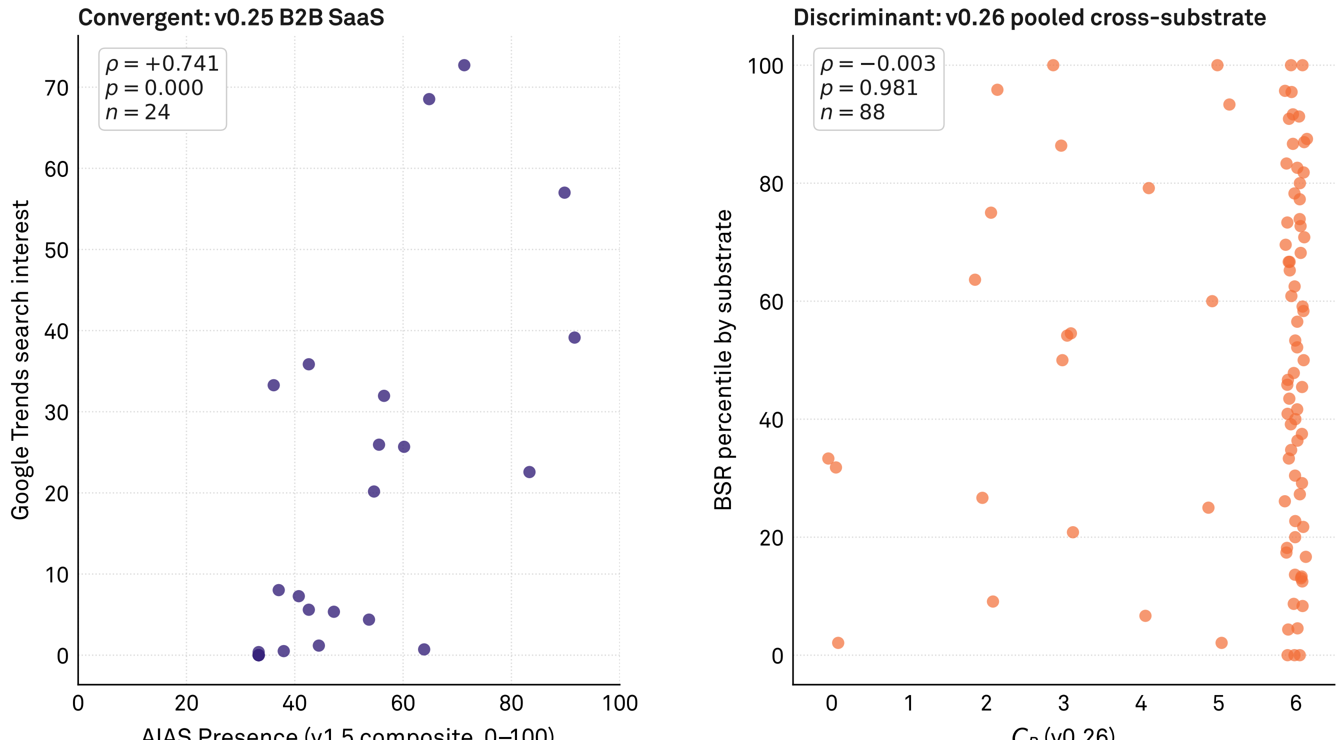
FINDING 03

Convergent + discriminant = construct validity

Convergent + discriminant: the boundary of AI Presence

Same construct, opposing relationships with related vs unrelated outcomes

Left: v0.25 B2B SaaS (indigo). Right: v0.26 pooled, within-substrate BSR percentile (warm).



Pattern 3: v0.25 + v0.26 jointly satisfy Campbell–Fiske construct validity.

Source: AIAS™ v0.26 | Protocol v1.6 | Third System™

Figure 3 · Convergent + discriminant validity. Left panel: v0.25 B2B SaaS — AIAS Presence composite (v1.5) vs. Google Trends search interest, rho = +0.741 (convergent). Right panel: v0.26 pooled across 4 substrates — C_P (0–6) vs. within-substrate BSR percentile, rho ≈ 0 (discriminant). Together, the two studies establish that AI Presence converges with related salience measures and diverges from unrelated commercial outcomes.

The Campbell–Fiske (1959) framework requires two conditions for construct validity: a measure must correlate with theoretically related measures (convergent) and must not correlate with theoretically unrelated measures (discriminant).

v0.25 established convergent validity: C_P correlated strongly with Google Trends search interest on B2B SaaS (rho = 0.74, $p < 0.001$). Brands that consumers search for are also brands that language models recognize. Both measures tap an underlying brand-salience dimension.

This study establishes discriminant validity: C_P shows zero correlation with Amazon BSR (pooled rho = 0.000). Brands that consumers buy on Amazon are not the same brands that language models recognize.

Together, the two studies map the boundary of what AI Presence measures. It captures brand salience within AI knowledge systems — a dimension that converges with search behavior but diverges from purchasing behavior. This is the defining characteristic of a novel construct: it is specific enough to fail where it should fail.

FINDING 04

Different data-generating processes

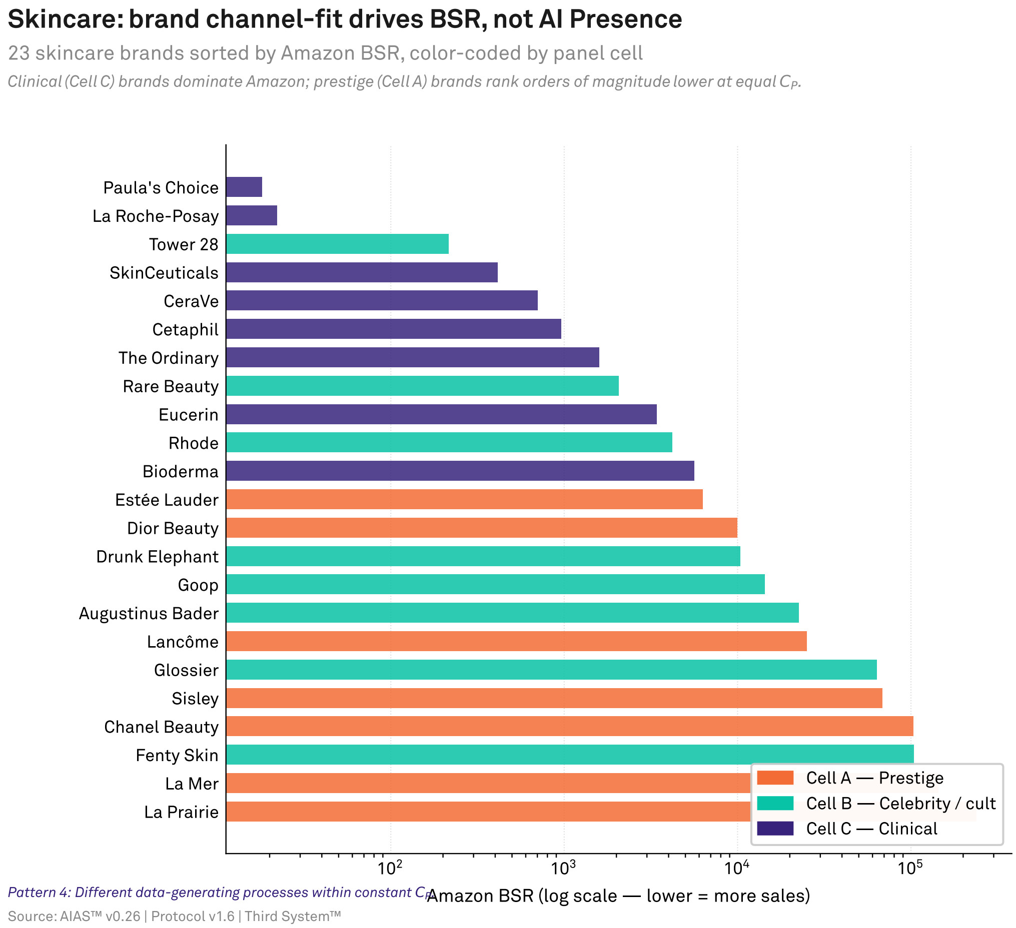


Figure 4 · Skincare: channel-fit drives BSR, not AI Presence. 23 skincare brands sorted by Amazon BSR, color-coded by panel cell. Clinical brands (Cell C — Paula’s Choice, La Roche-Posay, CeraVe) dominate Amazon; prestige brands (Cell A — La Prairie, La Mer, Sisley) rank 10,000× lower. The two cells span the same C_P range (predominantly 6), yet their BSR distributions barely overlap. Within constant AI Presence, channel-fit is the driver.

The null result is theoretically coherent. LLM training corpora overrepresent editorial, journalistic, and enthusiast discourse. A brand can be extensively discussed in knife forums, beauty publications, and audiophile communities (producing high C_P) without dominating Amazon sales, which reflects pricing, distribution logistics, Prime eligibility, advertising spend, and consumer purchasing inertia.

Consider the skincare substrate. La Roche-Posay (C_P = 6, BSR 22) and Paula’s Choice (C_P = 6, BSR 18) dominate Amazon with dermatologist-recommended, mass-accessible products. La Prairie (C_P = 6, BSR 237,462) is equally well-known to language models but ranks 10,000 times lower on Amazon — because it is a prestige brand that sells through department stores, not Amazon.

The signals originate in fundamentally different systems. AI Presence tracks what language models were trained on (editorial and encyclopedic content). Amazon BSR tracks what consumers actually purchase in a specific retail channel. These are independent dimensions of brand access, each requiring its own strategy and its own metrics. The Tri-System framework predicts exactly this: AI Availability, Mental Availability, and Physical Availability are three separate levers, not one.

Limitations

BSR captures sales velocity within one retail channel (Amazon), not total market performance. Brands with strong offline or DTC distribution may appear weaker in BSR than their true market position warrants. A multi-channel sales metric would provide a stronger discriminant validity test.

The v0.16 kitchen knives C_P was retrofitted under the v1.4+ protocol because the original v1.2 methodology did not produce comparable scores. The retrofit aligns the metric but introduces a provenance asymmetry documented in DEVIATIONS Entry 0.

Three testable substrates is the minimum for cross-substrate generalization. The v0.17 exclusion and v0.21 ceiling effect were not anticipated at pre-registration.

What’s next

Extend discriminant validity testing to additional commercial outcome measures: Sephora rankings, B&H; Photo bestsellers, and specialty retailer data. Prioritize behavioral correlates (click-through rates, recommendation acceptance) as convergent measures that sit closer to the theorized mechanism of AI Availability. Expand the Campbell–Fiske matrix with additional convergent measures (Wikipedia pageviews, news-corpus brand mentions) and discriminant measures (brand equity surveys, stock price).

Hypothesis scoring

Four pre-registered propositions tested across the v0.26 cross-substrate panel (kitchen knives, audiophile headphones, skincare, cosmetics).

H	Pre-registered prediction	Result	Status
P1	Per-substrate predictive validity	0 of 3 substrates met the rho [illegible] −0.40 threshold.	FALSIFIED
P2	Cross-substrate pooled correlation	Pooled rho = −0.0002, p = 0.998. Indistinguishable from zero.	FALSIFIED
P3	Cell A separation	Median BSR percentile: Cell A = 51.09, Other = 54.00. Negligible and non-significant.	FALSIFIED
P4	Absence–Presence alignment	Only 2 brands Amazon-absent (Nogent, Clé de Peau Beauté). Below the pre-registered minimum of 5 for inferential testing.	UNDETERMINED

Detailed hypothesis results

P1 — Per-substrate predictive validity: FALSIFIED Kitchen knives: rho = +0.048, p = 0.819 (n = 25) Audiophile headphones: rho = −0.116, p = 0.668 (n = 16) Skincare: rho = +0.049, p = 0.823 (n = 23) Cosmetics: ceiling effect (all C_P = 6), rho undefined

P2 — Pooled correlation: FALSIFIED 88 listed brands, percentile-normalized BSR: rho = −0.0002, p = 0.998

P3 — Cell A separation: FALSIFIED C_P [illegible] 4 (n = 78): median BSR percentile = 51.09 $C_P < 4$ (n = 10): median BSR percentile = 54.00 Mann-Whitney U: not significant

P4 — Absence–Presence alignment: UNDETERMINED Only 2 absent brands across all substrates. Insufficient sample for the pre-registered Mann-Whitney U test (minimum n = 5).

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Third System™ (research entity)

Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.6). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

González Castro, P. U. (2026). ***What AI Presence Does Not Predict: Amazon Best Sellers Rank as Discriminant Validity Evidence for the AIAS Construct (v0.26)***. Third System. thirdsystem.ai/v26 (SSRN 6847678)

Underlying datasets

BSR data: osf.io/ec6wh/v26/data/v26_bsr_master.csv

C_P retrofit: osf.io/ec6wh/v26/data/v16_cp_retrofit_aggregated.csv

Scoring verdicts: osf.io/ec6wh/v26/v26_verdicts.json

Methodology log: v1.6 (SSRN 6816340).

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